The Moving Target of Data Curation

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Abstract

This submission is for proposing a **tutorial.**

Sub-area keywords: Machine Learning, Data Mining and Knowledge Management, Data-Centric AI, Synthetic Data, Data Curation

Suggested Duration: quarter day

A myriad of works have shown that the data ingested by models are the ultimate driver of model capability and reliability; the focus is to maximize the quality of data. Still, data curation is scattered across big fields such as data efficiency and synthetic data generation. We propose this tutorial as a way to bring these two ideas together and show them in a unified light: that data *selection* and data *synthesis* are two sides of the same informativeness-maximization objective. Our tutorial is divided in three parts. **First**, we will overview why data curation is *still* a problem in 2027. Mainly, it is due to a change in demands from what models should learn from data. Among these demands, our data should change based on the training stage. Hence, our **second** part will cover how data curation differs in pre/mid-training, supervised fine-tuning, and post-training/reinforcement learning with verifiable rewards (RLVR). Finally, our **third** part will cover the technical details of data curation: the old techniques that are still relevant, as well as the newest, state-of-the-art methods in data curation. We'll conclude the tutorial with open problems for future research. This tutorial hopes to serve as an introduction to data curation research for novices, a refresher on newer topics for interested folks, and a foundational discussion for data curation experts.

Goal/Motivation of the Tutorial

Data curation has been studied for a very long time: from active learning, to data augmentation and data mixtures, to synthetic data generation. So why is it still a problem in 2027? What is it about generative models that makes our existing data sources insufficient? What does data curation research actually look like nowadays? How has the field changed with multi-modal generative models? These are the questions our tutorial aims to answer. The topic is timely for AAAI, given that naive scaling has saturated its benefits. Researchers have shown that the supply of high-quality human-generated text is itself running out, and that the next stage of progress will be driven by the *quality*—rather than quantity—of the data models ingest (Villalobos et al. 2024). We intend the tutorial to serve novices seeking an introduction into data curation, researchers catching up on recent work, and experts looking to help shape the field's direction.

Compared to our tutorial proposal, previous tutorials and surveys diverge by one of two ways:

1. They are more focused on a singular topic: either data efficiency (Wang et al. 2024b; Zhang et al. 2025; Qin et al. 2024; Albalak et al. 2024) or synthetic data generation (Liu et al. 2024; Wang et al. 2024a; Long et al. 2024).
2. They cover more of the methodology rather than the motivation or direction (Luo et al. 2025; Rao et al. 2025; Wu 2025).

Tutorial outline

Below is the tentative schedule for the tutorial.

Table 1: Tentative tutorial schedule.

Time	Content
0:00–0:10	(Dilek Hakkani-Tür) Motivation, background, and tutorial roadmap.
0:10–0:25	Part I (Dilek Hakkani-Tür): Why is data curation *still* unsolved? The new demand for training data that is dynamic, weakness-adaptive, and generalization-enabling.
0:25–0:42	Part II (Ishika Agarwal): How do curation goals differ across training stages.
0:42–0:50	Break for questions.
0:50–1:20	Part III (Ishika Agarwal): The major solution types and the shared informativeness signals.
1:20–1:35	Part IV (Arkajyoti Chakraborty): Data curation trends across downstream applications.
1:35–1:45	(Andreas Stolcke) Current gaps, future directions, and closing discussion.

History

Below we list the most closely related tutorials and explain how the proposed tutorial complements rather than duplicates them. To our knowledge, each prior tutorial addresses

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either selection or synthesis in isolation, or restricts their focus to a single sub-problem (attribution, efficiency, privacy). None unifies data selection and data synthesis under a shared informativeness principle, which is the contribution we aim to make.

- NeurIPS 2024 tutorial on "Opening the Language Model Pipeline" (Lo et al.). Difference: they focus on the broader topic of building pipelines for model development, rather than an in-depth view on data curation.
- NeurIPS 2024 tutorial on "Advancing Data Selection for Foundation Models" (Wang, Jia, Schmidt). Difference: they focus on data selection only, which is one half of data curation.
- ICML 2024 tutorial on "Foundations of Data-efficient Machine Learning" (Joshi, Mirzasoleiman). Difference: it focuses on data efficiency and does not treat curation as a whole[1].
- ICML 2024 tutorial on "Data Attribution at Scale" (Madry, Ilyas, Engstrom, Park, Georgiev). Difference: they focus on data attribution only, which is one signal for selection/curation.
- ICML 2025 tutorial on "DP-fy your DATA" (Ponomareva et al.). Difference: they approach synthetic data generation from a privacy and security angle, an overly narrow view from what we aim.

Target audience and prerequisites

Estimated number of participants: roughly 50-60.

We hope our tutorial serves three kinds of communities: (1) novices who are looking to understand what data curation research is, (2) researchers/practitioners who are trying to catch up on the latest in data curation research, and (3) experts who want to help shape the path for data curation research. We are hoping for around 50-60 people that roughly divide into these three categories.

The tutorial assumes familiarity with the basics of machine learning, and terms such as "pre-training", "supervised fine-tuning", and "reinforcement learning". The tutorial will be self-contained.

Content

By the end of the tutorial, attendees will be able to:

- Articulate **why data curation remains unsolved** in 2027 – the new demand that training data be dynamic, weakness-adaptive, and generalization-enabling (Part I);
- Understand **how curation goals differ** across pre-training, supervised fine-tuning, and RLVR, and select stage-appropriate strategies (Part II);
- Navigate the major solution types, and recognize the shared informativeness signals that unifies data selection and synthesis (Part III).
- Identify the **data curation trends in varying downstream** applications (Part IV):
  - In multi-modal generative models: text, image, video, speech (among others)
  - In training objectives: continual learning, encouraging fairness/mitigating under-representation, constructing data mixtures, alignment (among others)

Part I In this section, we will go over how the demands from generative models have changed. For example, during the inception of language generative models, the data needed to be simply large and clean (Longpre et al. 2024). After that, language models needed to follow instructions, so the data had to be `<instruction, input, output>` format, as well as be clean and logical (Zhou et al. 2023). Data also needed to be diverse (Jung et al. 2025) or representative (San Joaquin et al. 2024). Now, data needs to have reasoning chains (Jin et al. 2026), it needs to be curated for particular downstream tasks (Liu, Karbasi, and Rekatsinas 2024), and/or has to be on-policy (Zhang, Dai, and Peng 2025; Song and Zheng 2026). The same datasets that we used are not good enough to train today's models.

Part II The properties of data that each fine-tuning stage requires is different. For example, in the pre-training era, researchers want large and complete training data (Kim et al. 2026; Kang et al. 2026). In the supervised fine-tuning era, they require high quality labels (Bhatt et al. 2024; Harada et al. 2025). Currently, in the RLVR era, data needs to be of medium difficulty to avoid the zero-advantage problem (Wang et al. 2025; Li et al. 2026). We will go over all of these training regimes, and how they have discovered these patterns and properties.

Part III Data curation works can be divided into a bucket of solution types. Examples of solution types we want to cover, among others, are distribution matching (matching the distribution of your training set to a reference set), self-improvement (using data to improve a model by itself), influence estimation (calculating the impact of a data sample on a model), and in-context curation (carefully curating in-context examples to improve performance). We also want to talk about some of the principles that make data *selection* and data *synthesis* the same: how do researchers measure informativeness in both areas? What are the similarities? What are the differences, and why do they exist? This section will be our main technical section because it introduces our hypothesis that data selection and data synthesis are considered different fields, but in fact have much more in common than we think.

Part IV Data can be used for a variety of downstream applications. They can be used in a variety of modalities (text, image, video, speech), and each modality comes with their own challenges. We want to overview the common concepts and challenges when thinking of data curation for these modalities. Data can also be used to introduce a variety of capabilities: learning without forgetting, upsampling minority classes without making the data unrealistic, understanding how to combine a variety of datasets depending on model

[1]Note: our co-presenter Mirzasoleiman led this tutorial in ICML; we build off of her tutorial by talking about the full selection–synthesis spectrum and the latest post-training literature.

capability, etc. We will broadly cover the trends in these domains to get a picture of how dynamic data curation affects these problems and how researchers are solving them.

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